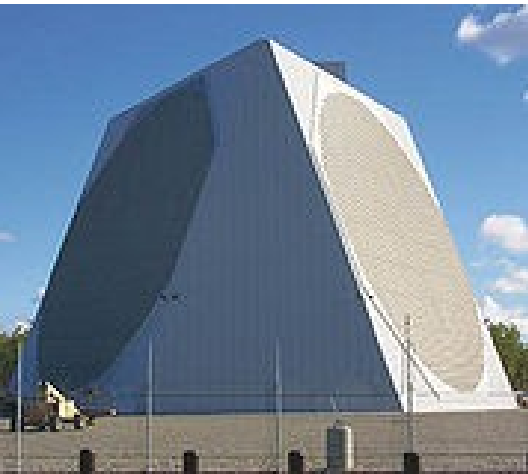


ECE 3317

Dr. Jiefu Chen

Notes 1

Introduction



Adapted from notes by Prof. Stuart A. Long

Why study electromagnetic waves?

Motivation:

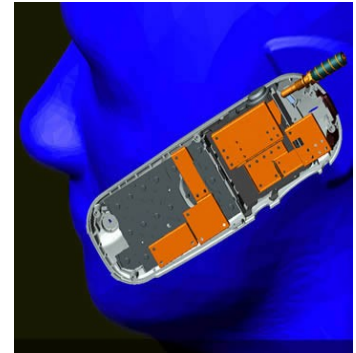
Most basic of all ECE courses: Electromagnetics provides the foundation for all electrical sciences and engineering. All of circuit theory is a special case (low frequency, where dimensions are small relative to a wavelength).

Electromagnetics explains physical phenomena: What is light? How do electric and magnetic fields behave?

Important to know about EM: Extremely important for areas such as antenna design, microwave engineering, RF design. Also very important for power engineering and micro/nano-electronics. It is good to know for all areas of ECE.

Applications

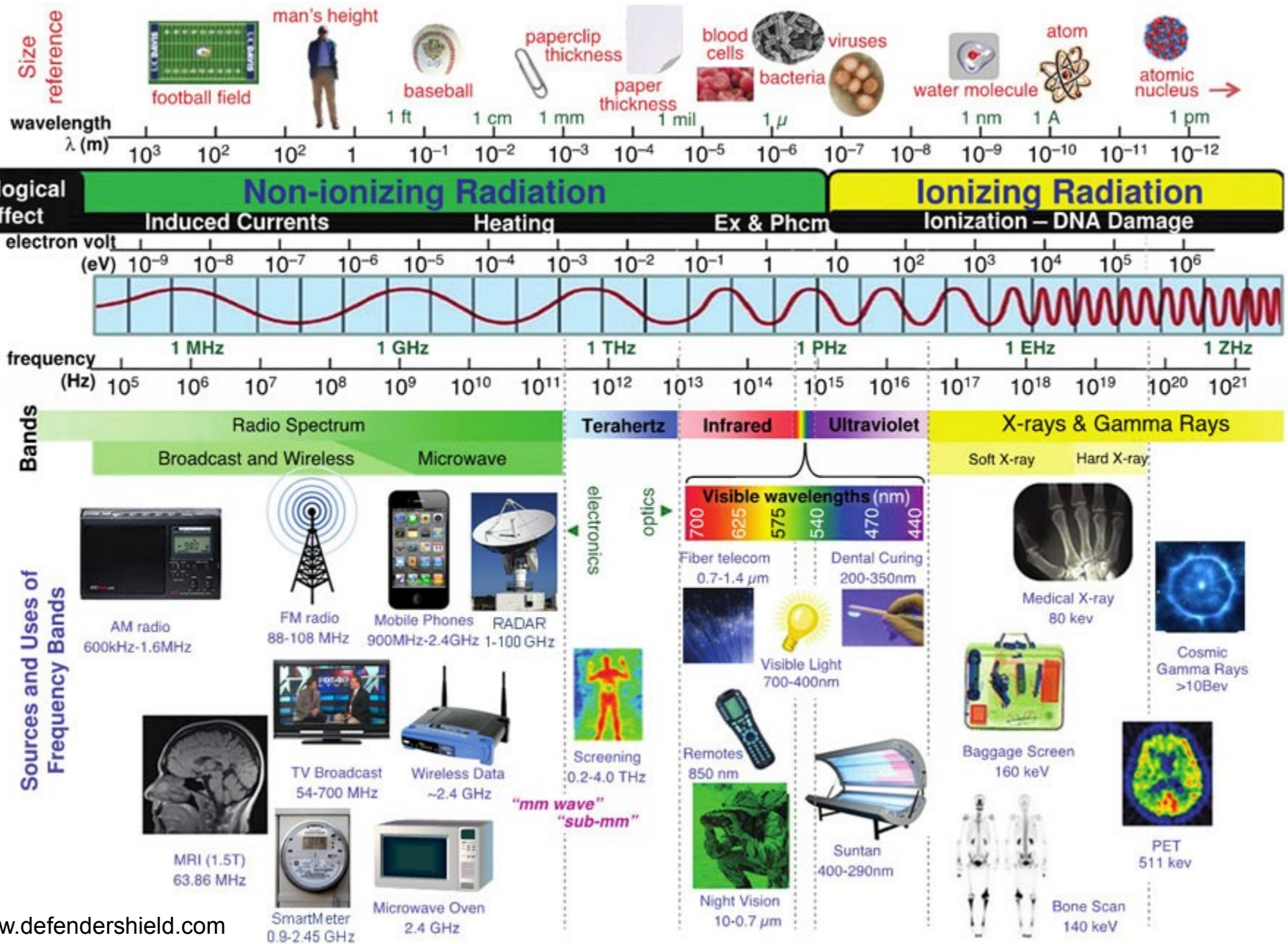
Applied Electromagnetics: Antennas, Radar, Microwaves, RF circuits.



Wireless Communications: Cell phones; Bluetooth; Wireless routers; Cordless handsets; RFIDs.



Electromagnetic Spectrum



Microwave Oven (2.45 GHz)

www.rentfuerte.com



- How microwave cooking works?
- Why sometimes microwave heats food unevenly?

Millimeter Wave Imaging (30-300 GHz)

en.wikipedia.org/wiki/Full_body_scanner

www.nist.gov



- Why millimeter wave sees through clothing?
- Is millimeter wave scanning safe (for pregnant women)?